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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 20

Network Security

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COMPUTER · CHAPTER 20

Network Security

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IN THIS CHAPTER

- Network Security
- Types Of Network Security Devices
- Comparison Of Security Devices
- Firewall
- How A Firewall Works
- Firewall Rule Table

NETWORK SECURITY

Network Security refers to the policies, technologies, devices, and practices used to protect computer networks, data, and resources from unauthorized access, misuse, modification, or cyber attacks.

Objectives of Network Security

- Protect Data
- Prevent Unauthorized Access
- Detect Attacks
- Ensure Safe Communication
- Maintain Network Availability

Exam Point

✔ Network Security protects both hardware and software resources connected to a network.

TYPES OF NETWORK SECURITY DEVICES

Network security devices are used to monitor, detect, prevent, and respond to security threats.

Network Security Devices

```
|
├── Active Devices
├── Passive Devices
├── Preventative Devices
└── Unified Threat Management (UTM)
```

1. ACTIVE DEVICES

Active security devices actively monitor network traffic and block malicious or unwanted traffic.

✔ Active Devices = Detect + Block

2. PASSIVE DEVICE

Passive devices monitor network activities and report suspicious behavior but do not directly block attacks.

✔ Passive Devices = Detect + Report

3. PREVENTATIVE DEVICES

Preventative devices identify vulnerabilities before attackers exploit them.

✔ Preventative Devices help prevent attacks before they occur.

4. UNIFIED THREAT MANAGEMENT (UTM)

UTM (Unified Threat Management) is an all-in-one security solution that combines multiple security features into a single device.

✔ UTM = Multiple security functions in one device.

COMPARISON OF SECURITY DEVICES

Device Type	Function	Example
Active	Detect and Block	Firewall
Passive	Detect and Report	IDS
Preventative	Identify Weaknesses	Penetration Testing Tool
UTM	All-in-One Security	Security Gateway

FIREWALL

A Firewall is a network security system that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a barrier between:

Trusted Network

|

Firewall

|

Untrusted Network (Internet)

✔ Firewall is the first line of defense in network security.

The firewall checks each packet against predefined rules and decides whether to:

- Allow
- Reject
- Block
- Log the traffic.

HOW A FIREWALL WORKS

Internet Traffic



FIREWALL RULE TABLE

Firewalls maintain rule tables that determine:

- Which traffic is allowed
- Which traffic is blocked

TYPES OF FIREWALLS

A. HARDWARE FIREWALL

A Hardware Firewall is a physical device installed between a network and the Internet.

Advantages

- Protects entire network
- High performance
- Difficult to bypass

Exam Point

✔ Hardware Firewall protects the whole network.

B. SOFTWARE FIREWALL

A Software Firewall is installed on an individual computer.

Characteristics

- Installed as software
- Protects one device
- Monitors local applications

Examples

- Windows Defender Firewall
- Personal Firewall Software

Exam Point

✔ Software Firewall protects individual computers.

HOST-BASED FIREWALL

A firewall installed on every device (host) in a network.

Example

Windows Firewall on a laptop.

NETWORK-BASED FIREWALL

A firewall installed at the network boundary, usually on servers or routers.

Characteristics

- Protects the entire network

Example

Firewall integrated into enterprise routers.

ANTIVIRUS

Antivirus software is a security tool used to detect, prevent, and remove malware from computer systems.

Originally designed to remove viruses, modern antivirus programs provide protection against many types of cyber threats.

Protected Against

- Viruses
- Worms
- Trojan Horses
- Spyware
- Adware
- Keyloggers
- Ransomware
- Phishing Attacks

Exam Point

✔ Antivirus protects against multiple types of malware, not just viruses.

FAMOUS ANTIVIRUS SOFTWARE

- Norton AntiVirus
- McAfee
- Quick Heal
- Kaspersky Anti-Virus
- Avast Antivirus

CONTENT FILTERING

Content Filtering is the process of blocking inappropriate, harmful, or unauthorized content from reaching users.

Uses

- Web Filtering
- Email Filtering
- Spam Blocking
- Adult Content Blocking

Example

When a user tries to access a restricted website:

ACCESS DENIED message appears.

TYPES OF CONTENT FILTERING

A. WEB FILTERING

Blocks unwanted websites.

Examples

- Gambling Sites
- Adult Websites
- Malicious Websites

B. EMAIL FILTERING

Filters unwanted emails.

Examples

- Spam Emails
- Phishing Emails
- Malicious Attachments

Exam Point

✔ Content Filtering helps reduce spam and unauthorized access.

INTRUSION DETECTION SYSTEM (IDS)

An Intrusion Detection System (IDS) is a security system that monitors network traffic for suspicious activities and generates alerts.

Exam Point

✔ IDS mainly detects and reports attacks.

INTRUSION PREVENTION SYSTEM (IPS)

An Intrusion Prevention System not only detects attacks but also blocks them automatically.

Functions

- Detect
- Block
- Prevent
- Log

Exam Point

✔ IPS = IDS + Blocking Capability

QUICK REVISION NOTES

Security Devices

- Active → Detect + Block
- Passive → Detect + Report
- Preventative → Find Vulnerabilities
- UTM → All-in-One Security

Firewalls

- Controls network traffic
- Uses predefined rules
- Acts as security barrier

Antivirus

- Detects and removes malware

Content Filtering

- Blocks unwanted websites and emails

IDS

- Detects attacks

IPS

- Detects and blocks attacks

MOST IMPORTANT FACTS FOR EXAMS

1. Firewall acts as a barrier between trusted and untrusted networks.

2. Hardware Firewall protects the entire network.

3. Software Firewall protects individual computers.

4. Host-Based Firewall is installed on each device.

5. Network-Based Firewall is installed on servers or routers.

6. Antivirus detects and removes malware.

7. Modern antivirus protects against viruses, worms, Trojans, spyware, and ransomware.

8. Content Filtering blocks unwanted websites and emails.

9. IDS detects and reports attacks.

10. IPS detects and prevents attacks.

11. Passive devices only monitor and report.

12. Active devices detect and block threats.

13. UTM combines multiple security functions into one device.

14. Firewalls operate using rule tables.

15. Norton and McAfee are popular antivirus software